

## DataBase Overview and History

In Software Data could Be Anything It may be User Data, Product Data or It Could Be a Raw Information or It would Be a fact.

So Data is highly Important and knowing about the Data is Equally Important In Any stream.

So knowing about Data & Data Bases is Essential In Software Industry.

## History of DataBases

In 1960 the data is stored in file Based System.

For Example

Product.txt

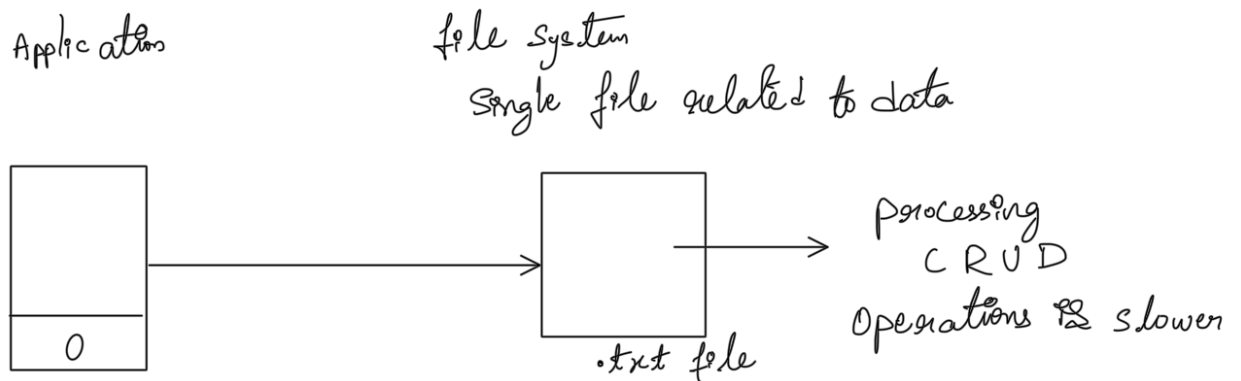
| ProductID | Pname | price | customername | Qty | orderID | shipment |
|-----------|-------|-------|--------------|-----|---------|----------|
| 1         | Pen   | 10    | Veesh        | 1pc | 1       | 111      |
| 2         | Sugar | 40    | Veesh        | 1kg | 1       | 121      |
| 3         | Salt  | 10    | Veesh        | 1kg | 1       | 222      |
| 4         | Paper | 70    | Veesh        | 1pc | 1       | 123      |

Data Duplication → If The Name is Repeated Again & Again then It stored repeatedly

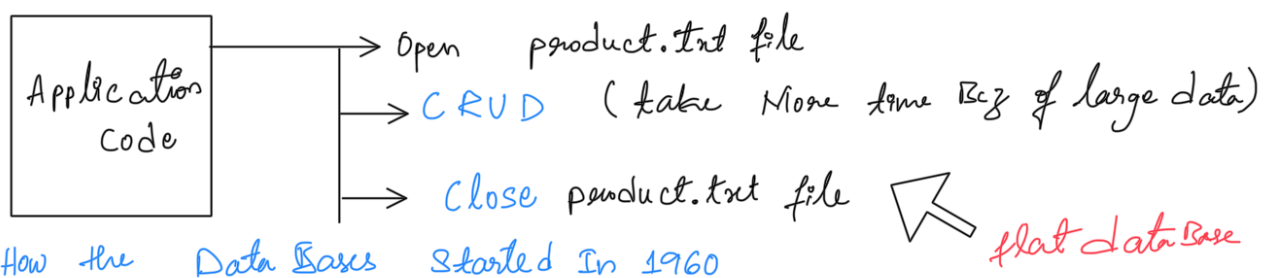
flat DataBase

All the Data is stored In Single text file, Imagine How large the file size Becomes if Information Related to the Data Increases.

Everytime we have to open the file and create a New record, Update a record, Read a record, Delete a Record. How slow will be this operations if the Data is Very large.



The idea of Single file data storing increases complexity



It's How the Data Bases Started In 1960

The challenges faced during single file storage are

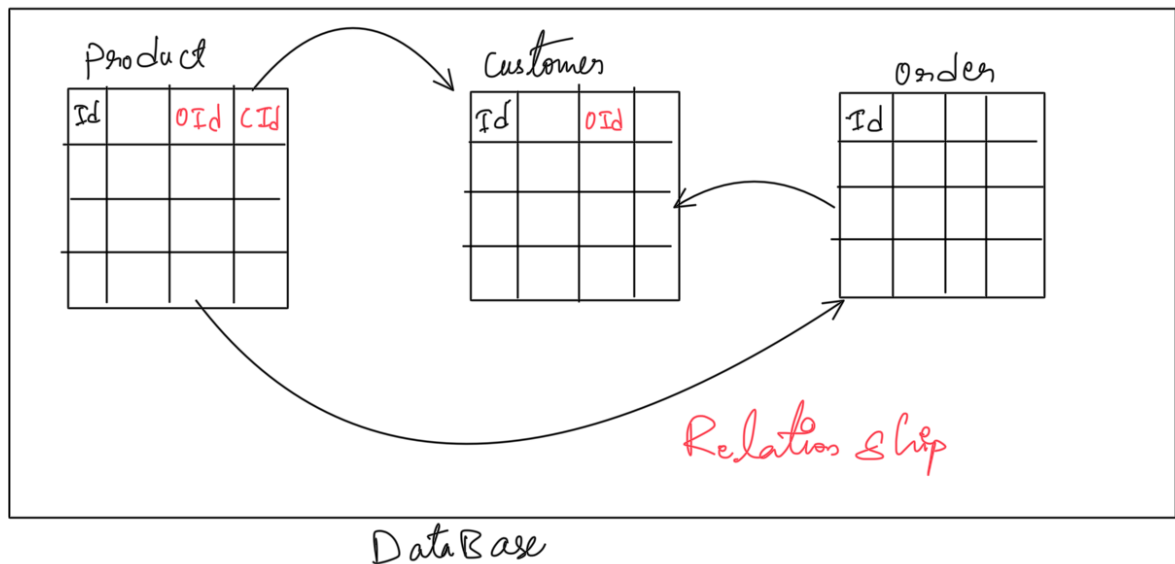
1. Data Duplication
2. Security Challenges (Bcz the data stored in local machine)
3. No standardization (any change in data schema directly Break the APP)
4. Building Relational Data is a Challenge (data was flat)

So In 1970 Data Bases revolution started

Relational Model & SQL Language Invented

- 1) Relational Establishment helped to manipulate only selective data instead of fetching all data and processing. Which is faster and smarter way for CRUD operations
- 2) Schema is introduced (Schema decides the properties or data types of each column in the database)
- 3) SQL language invention changed the entire database evolution

Relationship Establishment (We basically store data)



Data is distributed in tables which saves the space compared to earlier flat database.

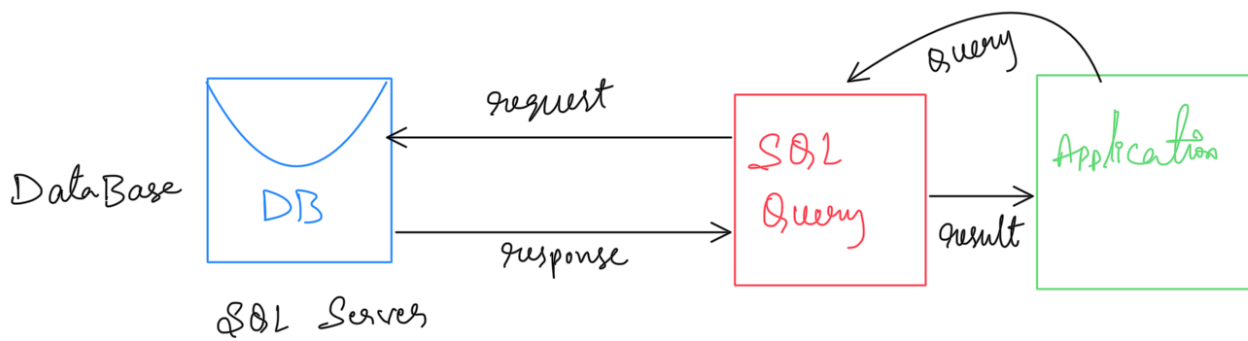
We can get related data through the joins

SQL (Structured Query Language)

SQL (The language that SQL Server understands)

SQL is a standard language used to store, retrieve, manage and manipulate data in a Relational Database.

SQL helps to talk with database

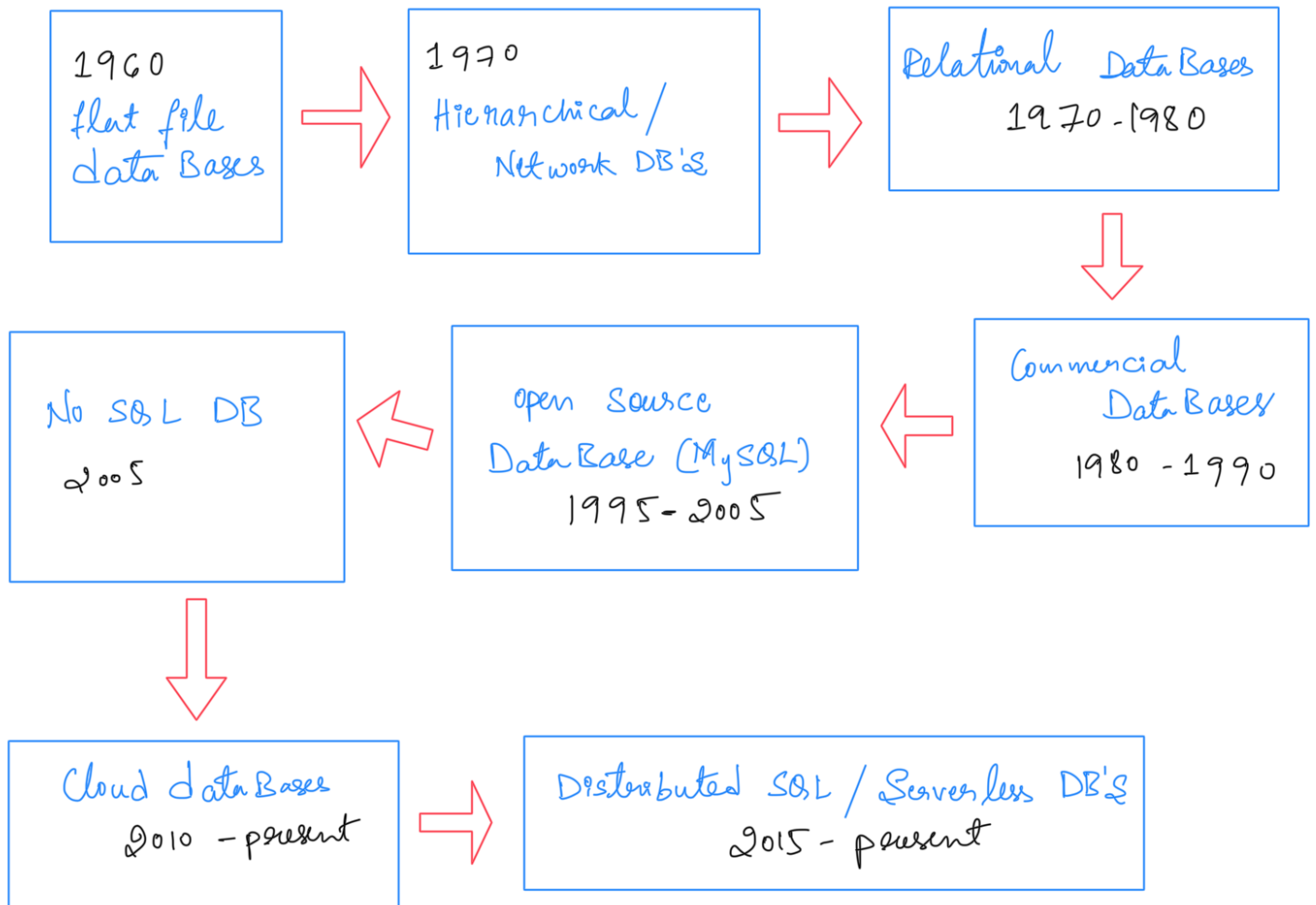


## What SQL Server does

- Stores data in **tables** (rows & columns)
- Uses **SQL** to query and manage data
- Ensures **data security, reliability, and performance**
- Supports **transactions, indexes, views, stored procedures, and triggers**

- ① SQL Server is a Data Base, have all the Data Which is related to a Application or something
- ② To access or perform CRUD operation on that Data we use SQL Query language
- ③ SQL Query language solved the issue of standardization in the flat data bases. Bcz programmers don't need to worry about the file name, path, server etc information
- ④ SQL Query is standardized

## Revolution of Data Bases.



### 🔥 How NoSQL Databases Support Horizontal Scaling

Horizontal Scaling = Scale out, not up

#### ◆ Key Techniques Used by NoSQL

##### 1 Sharding

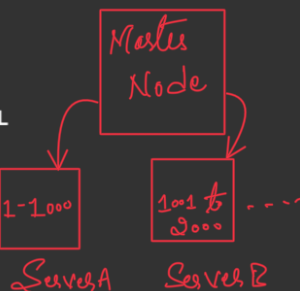
- Data split across multiple nodes
- Each node stores a portion of data

code

```
UserID 1-1000 → Server A  
UserID 1001-2000 → Server B
```

##### 2 Replication

- Same data copied across nodes
- Improves availability & fault tolerance



##### 3 Shared-Nothing Architecture

- Each node is independent
- No central bottleneck

##### 4 Eventual Consistency

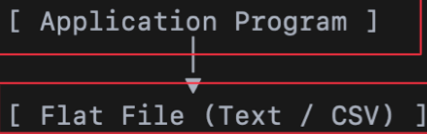
- Data syncs over time
- Improves performance & availability

##### 5 Automatic Rebalancing

- When new nodes added, data redistributes automatically

## 2. Flat File Database – Block Diagram

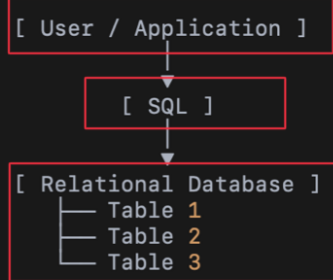
code



- ✓ No relations
- ✗ Data redundancy
- ✗ No security

## 3. Relational Database (RDBMS) – Block Diagram

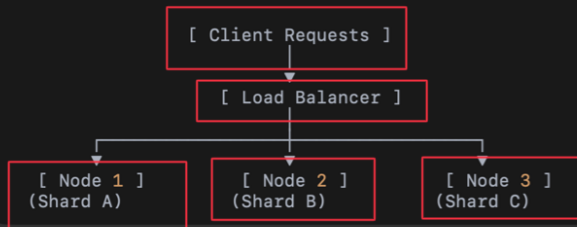
code



- ✓ Tables, Keys
- ✓ ACID properties
- ✗ Limited horizontal scaling

## 4. NoSQL Database – Horizontal Scaling Block Diagram ★

code

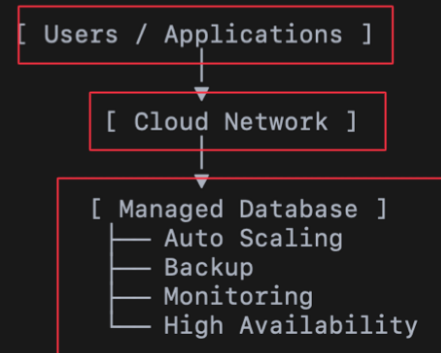


Key Points:

- Sharding → Data split across nodes
  - Replication → Data copied for fault tolerance
  - Shared-nothing architecture
- ✓ Horizontal scaling
  - ✓ High availability
  - ✓ Big data support

## 5. Cloud Database – Block Diagram

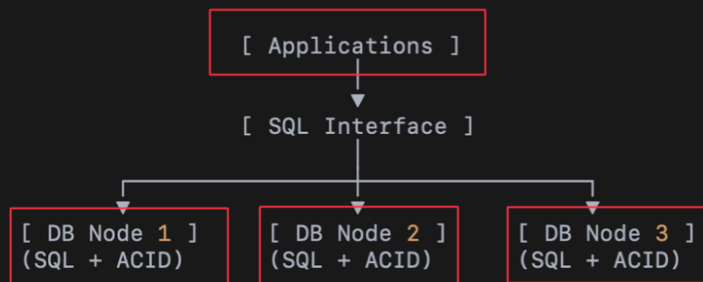
code



- ✓ Pay-as-you-use
- ✓ Fully managed
- ✓ Global access

## 6. Distributed SQL Database – Block Diagram

code

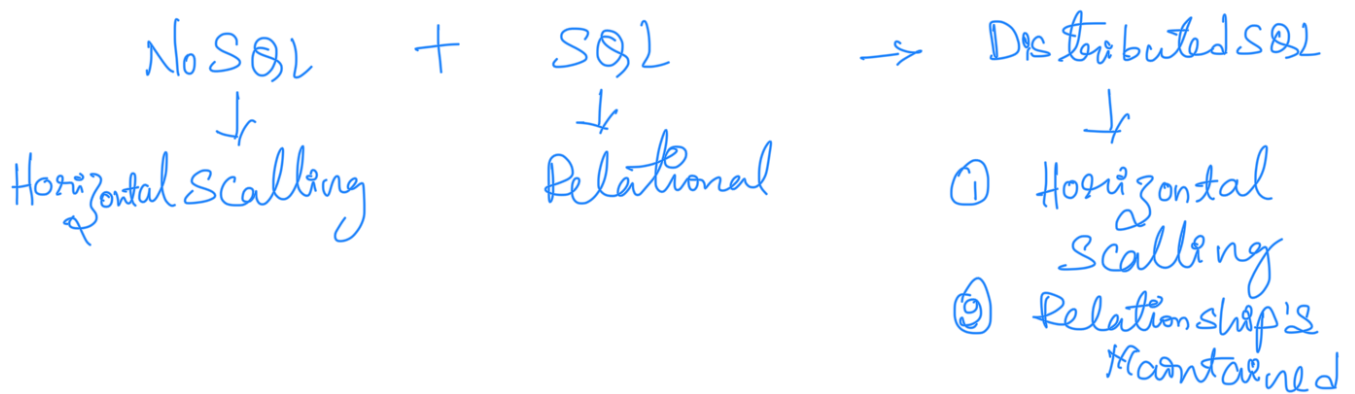


- ✓ SQL support
- ✓ ACID transactions
- ✓ Horizontal scaling like NoSQL

In NoSQL Databases data is distributed so it took away the capability of joins bcz data is distributed so joining is costly bcz data have to go through network calls.

But SQL data is not horizontally scalable.

Combination of features of SQL & NoSQL is Distributed SQL.



#### What it means

An exact query returns results only when the data exactly matches the given condition.

#### How it works

- Uses SQL conditions
- Keyword / value-based matching
- Deterministic (true or false)

#### Example (SQL)

```
sql
SELECT * FROM products
WHERE name = 'iPhone 14';
```

✓ Returns results only if name = 'iPhone 14'

✗ Will NOT return:

- iPhone 13
- iPhone 14 Pro
- Apple phone

#### Used in

- Relational databases (MySQL, SQL Server)
- Traditional NoSQL (key-value lookups)

#### SQL Examples

##### 1 Pattern matching using LIKE

```
sql
SELECT *
FROM students
WHERE name LIKE 'Veer%';
```

→ Matches: Veeresh, Veerendra, Veera

```
sql
SELECT *
FROM students
WHERE name LIKE '%esh';
```

##### 2 Multiple possibilities using IN

```
sql
SELECT *
FROM students
WHERE branch IN ('CSE', 'ISE', 'AI');
```

#### 2 Similar Query (AI / Intelligent Databases)

##### What it means

A similar query returns results that are conceptually or semantically related, not exactly the same.

##### How it works

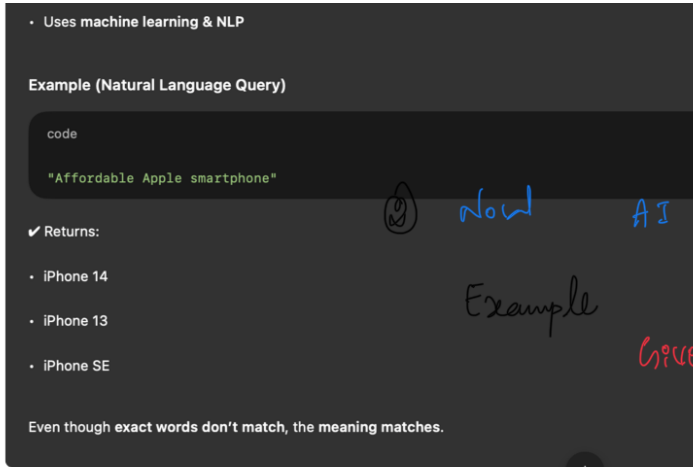
- Converts data into vectors (embeddings)
- Measures similarity distance (cosine, euclidean)

① Before

Query was Exact (Searching for exact Match)

Example

Select name from product where  
product name = " "



②

Now

AI Databases

Search

for

Similar Matches

Example

Give me document like "This Document"  
↓  
Similar